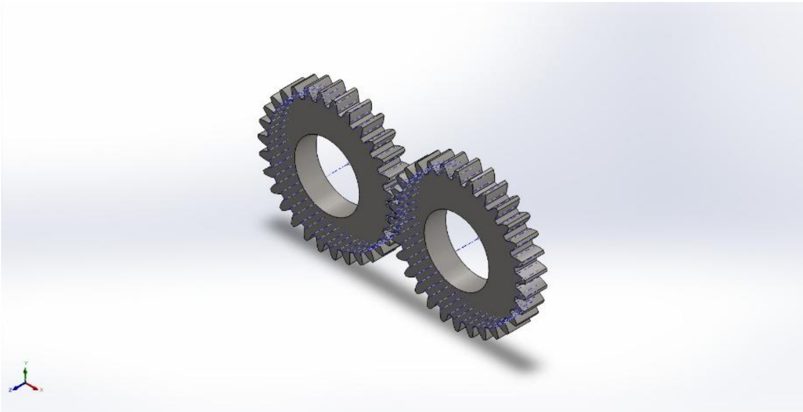
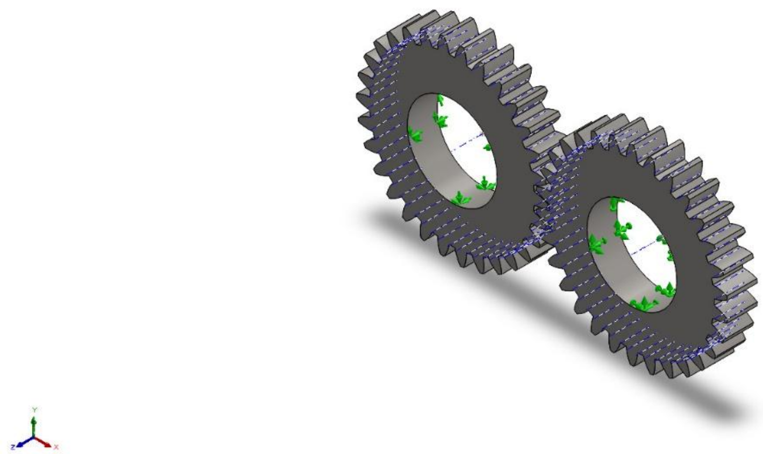


Simulation of Assem2

Date: 1 August 2025
Designer: Solidworks
Study name: Spur Gear
Analysis type: Static

Table of Contents

Model Information



Model name: Assem2
Current Configuration: Default

Solid Bodies

Document Name and Reference	Treated As	Volumetric Properties	Document Path/Date Modified
Cut-Extrude1 	Solid Body	**Mass:0.160666 kg Volume:2.05982e-05 m^3 Density:7,800 kg/m^3 Weight:1.57452 N**	**C:\Users\prati\AppData\Local\Temp\swx6660\VC~~\Assem2\Part1^Assem2.sldprt** Aug 10 02:18:31 2025
Cut-Extrude1 	Solid Body	**Mass:0.160666 kg Volume:2.05982e-05 m^3 Density:7,800 kg/m^3 Weight:1.57452 N**	**C:\Users\prati\AppData\Local\Temp\swx6660\VC~~\Assem2\Part1^Assem2.sldprt** Aug 10 02:18:31 2025

Study Properties

Study name	Spur Gear
Analysis type	Static
Mesh type	Solid Mesh
Thermal Effect:	On
Thermal option	Include temperature loads
Zero strain temperature	298 Kelvin
Include fluid pressure effects from SOLIDWORKS Flow Simulation	Off
Solver type	Automatic
Inplane Effect:	Off
Soft Spring:	Off
Inertial Relief:	Off
Incompatible bonding options	Automatic
Large displacement	Off
Compute free body forces	On
Friction	Off
Use Adaptive Method:	Off
Result folder	SOLIDWORKS document (c:\users\prati\appdata\local\temp)

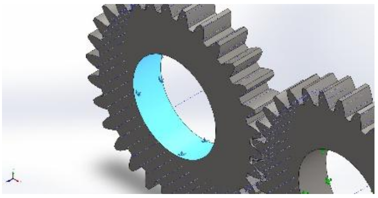
Units

Unit system:	SI (MKS)
Length/Displacement	mm
Temperature	Kelvin
Angular velocity	Rad/sec
Pressure/Stress	N/m^2

Material Properties

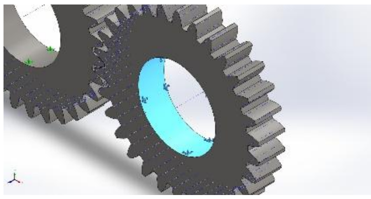
Model Reference	Properties	Components
	**Name:** Plain Carbon Steel **Model type:** Linear Elastic Isotropic **Default failure criterion:** Max von Mises Stress **Yield strength:** 2.20594e+08 N/m^2 **Tensile strength:** 3.99826e+08 N/m^2 **Elastic modulus:** 2.1e+11 N/m^2 **Poisson's ratio:** 0.28 **Mass density:** 7,800 kg/m^3 **Shear modulus:** 7.9e+10 N/m^2 **Thermal expansion coefficient:** 1.3e-05 /Kelvin	SolidBody 1(Cut-Extrude1)(Part1^Assem2-1), SolidBody 1(Cut-Extrude1)(Part1^Assem2-3)
Curve Data:N/A		

Loads and Fixtures

Fixture name	Fixture Image	Fixture Details
On Cylindrical Faces-1		**Entities:** 1 face(s) **Type:** On Cylindrical Faces **Translation:** 0, 0.087 rad., 0 **Units:** mm

Resultant Forces

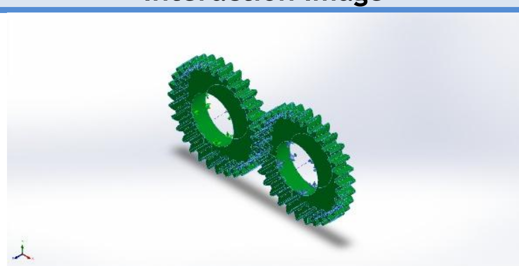
Components	X	Y	Z	Resultant
Reaction force(N)	254,365	585,249	6.02395	638,136
Reaction Moment(N.m)	0	0	0	0

Fixture name	Fixture Image	Fixture Details
Fixed-1		**Entities:** 1 face(s) **Type:** Fixed Geometry

Resultant Forces

Components	X	Y	Z	Resultant
Reaction force(N)	-254,365	-585,249	-6.02256	638,136
Reaction Moment(N.m)	0	0	0	0

Interaction Information

Interaction	Interaction Image	Interaction Properties
Component Interaction-1		**Type:** Contact (Surface to surface) **Components:** 2 Solid Body (s)

Mesh information

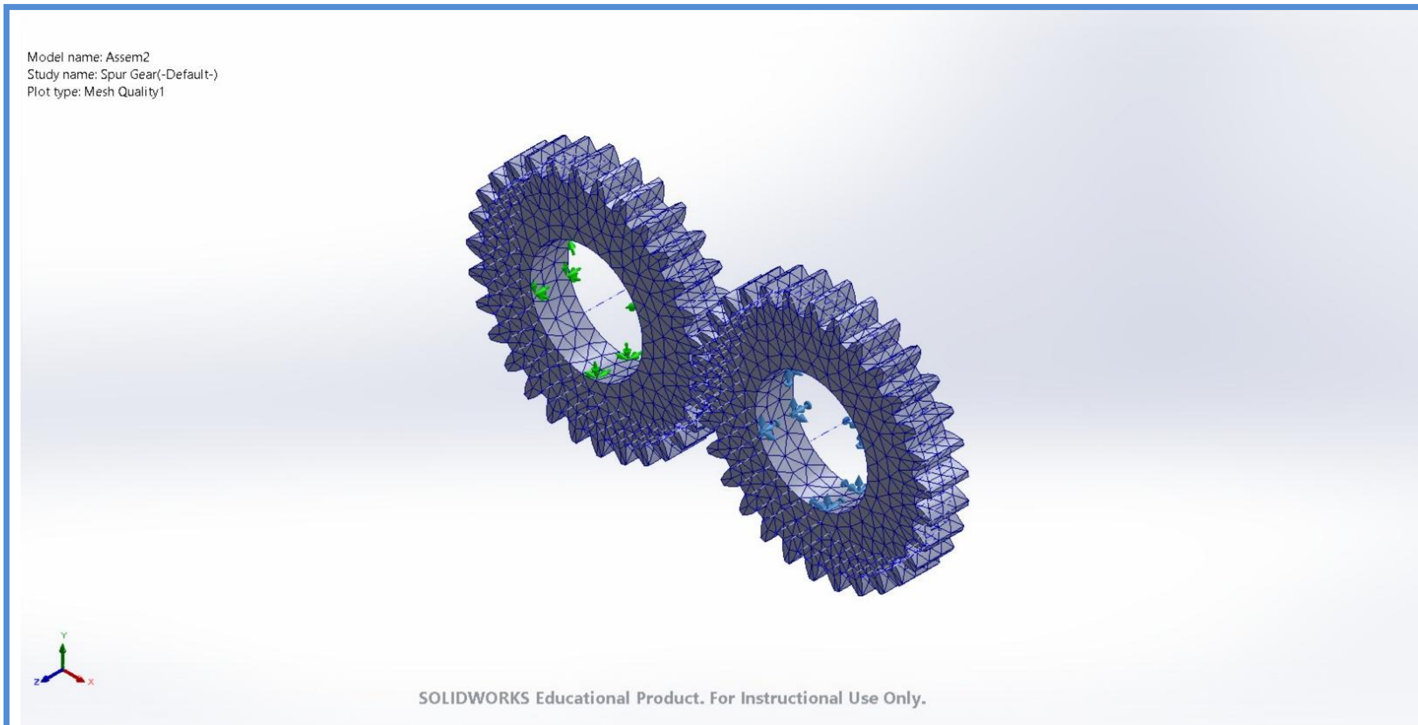
Mesh type	Solid Mesh
Mesher Used:	Blended curvature-based mesh
Jacobian points for High quality mesh	16 Points
Maximum element size	3.45488 mm
Minimum element size	3.45488 mm
Mesh Quality	High
Remesh failed parts independently	Off
Reuse mesh for identical parts in an assembly (Blended curvature-based mesher only)	Off

Mesh information - Details

Total Nodes	20984
Total Elements	12282
Maximum Aspect Ratio	3.5976
% of elements with Aspect Ratio < 3	96.7
Percentage of elements with Aspect Ratio > 10	0
Percentage of distorted elements	0
Time to complete mesh(hh;mm;ss):	00:00:01
Computer name:	MSI

Mesh Quality Plots

Name	Type	Min	Max
Quality1	Mesh	-	-



Assem2-Spur Gear-Quality-Quality1

Resultant Forces

Reaction forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	-0.0363479	0.423767	0.000937045	0.425324

Reaction Moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	0

Free body forces

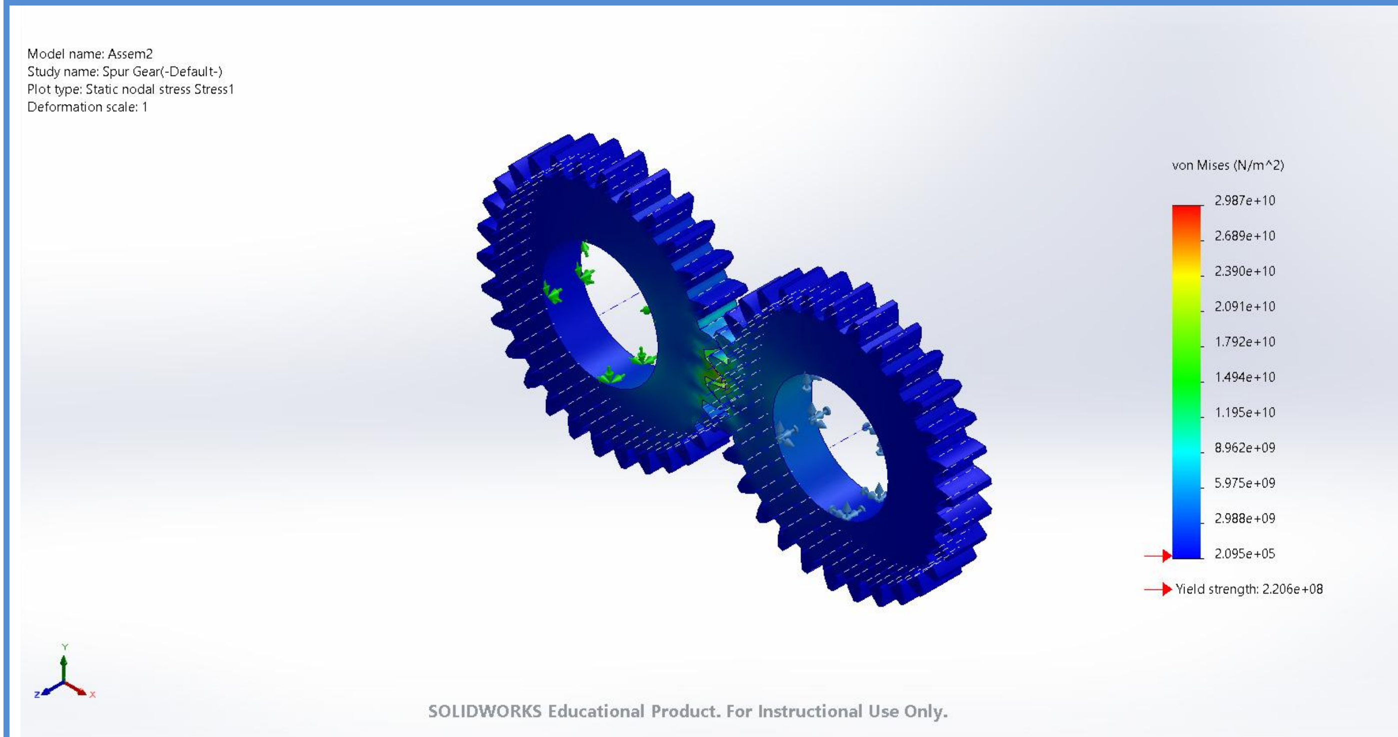
Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	-0.0675979	0.0516968	0.00213677	0.0851269

Free body moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	1e-33

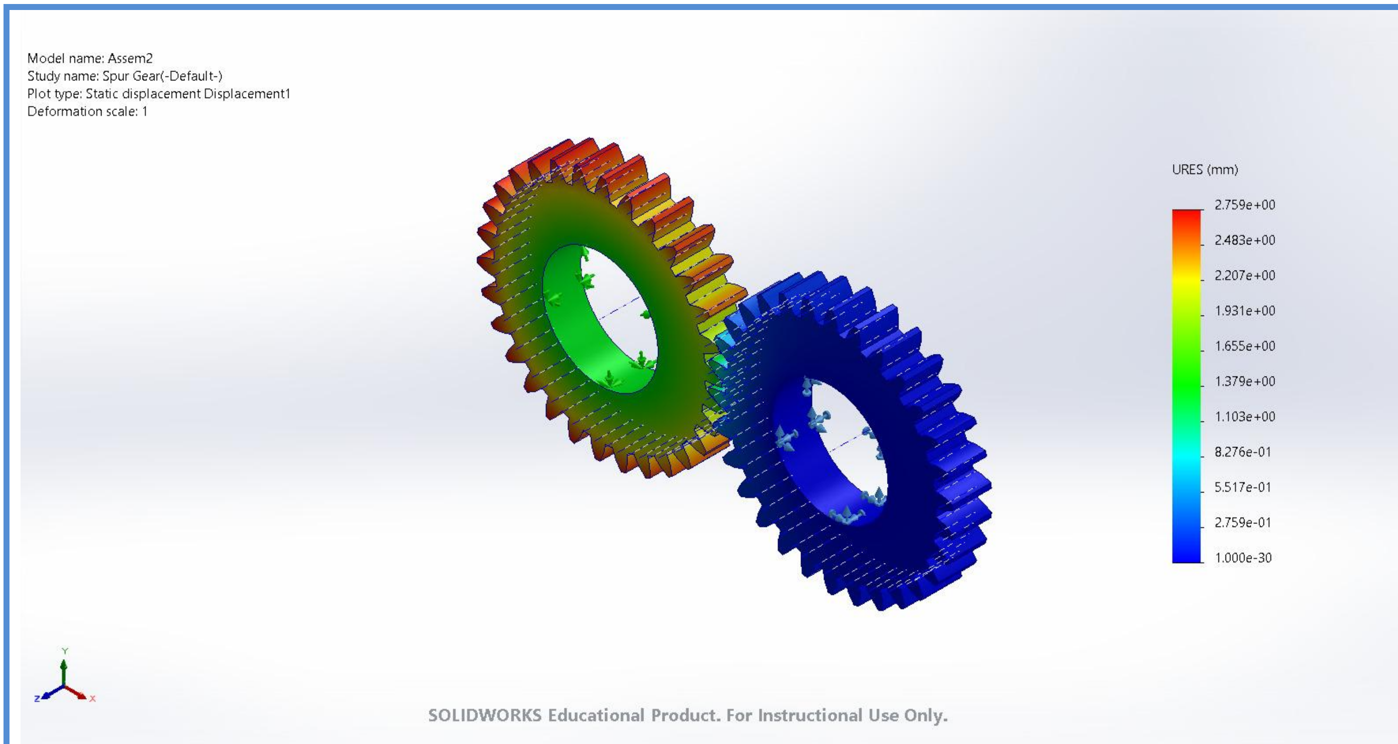
Study Results

Name	Type	Min	Max
Stress1	VON: von Mises Stress	2.095e+05N/m^2 Node: 8233	2.987e+10N/m^2 Node: 20669



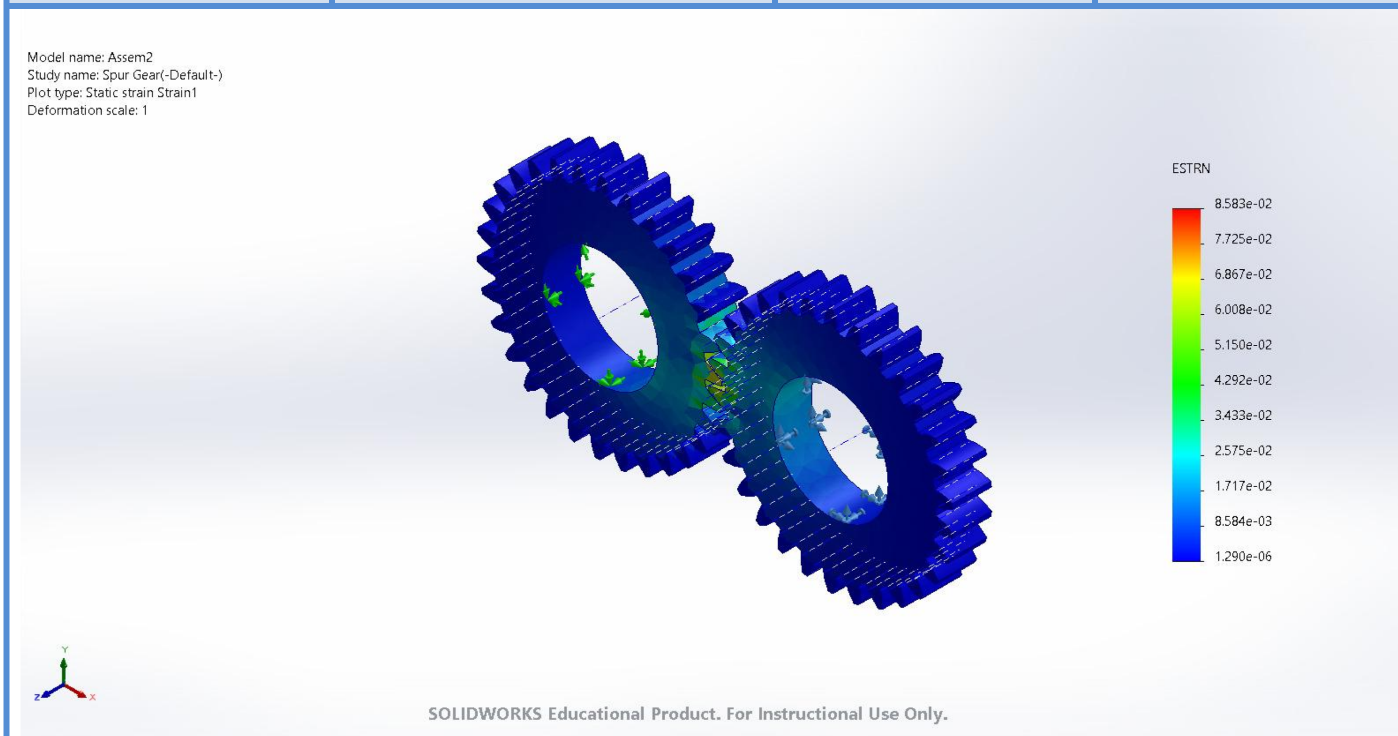
Assem2-Spur Gear-Stress-Stress1

Name	Type	Min	Max
Displacement1	URES: Resultant Displacement	0.000e+00mm Node: 10711	2.759e+00mm Node: 8278



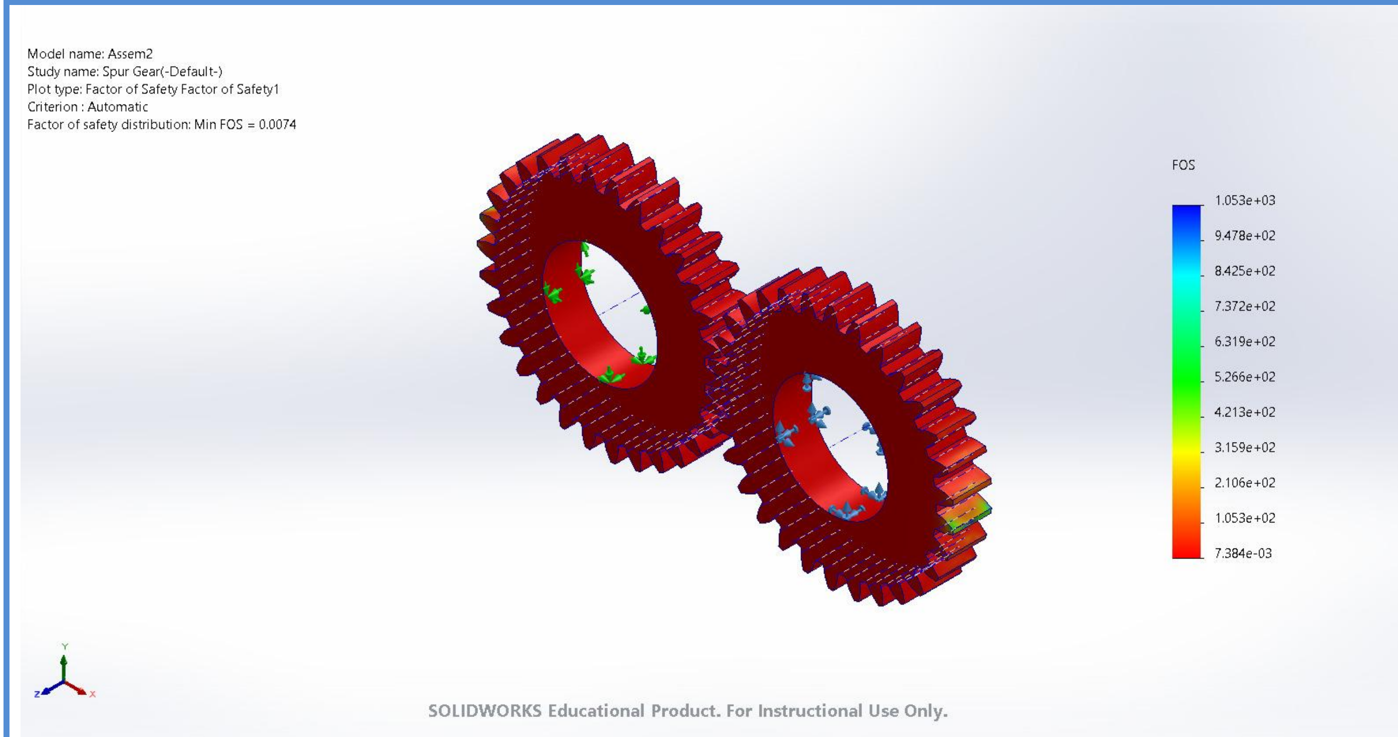
Assem2-Spur Gear-Displacement-Displacement1

Name	Type	Min	Max
Strain1	ESTRN: Equivalent Strain	1.290e-06 Element: 10556	8.583e-02 Element: 11521



Assem2-Spur Gear-Strain-Strain1

Name	Type	Min	Max
Factor of Safety1	Automatic	7.384e-03 Node: 20669	1.053e+03 Node: 8233



Assem2-Spur Gear-Factor of Safety-Factor of Safety1